Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination – 2018

EE1107 - ELECTRICAL PRINCIPLES – A

Answer Five question only Time : 2 hours

Q1 (Total 20 Marks)

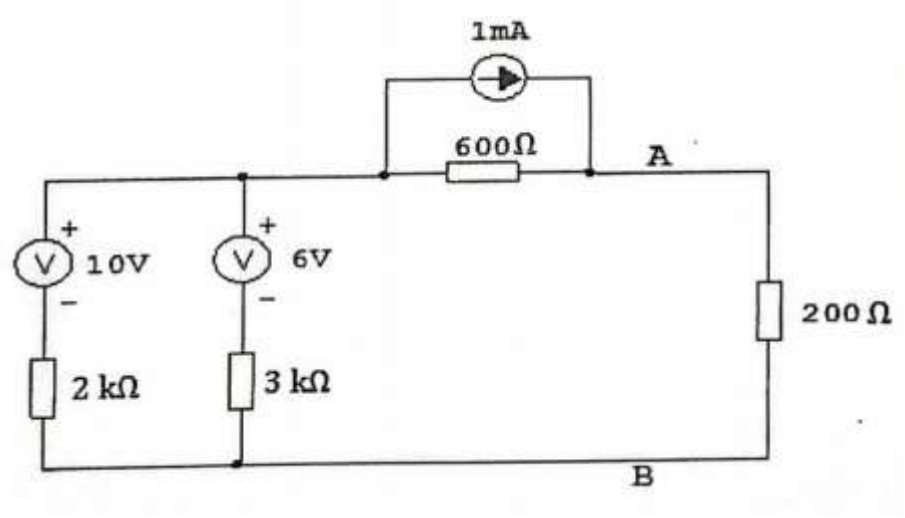
Figure-Q1 shows a d.c. closed loop network.

(a) Derive Thevenin equivalent circuit across AB terminal and calculate the current flowing in the 200Ω resistor.

(b) Determine Norton equivalent circuit across AB terminal and the current flowing in the 200Ω resistor.

(c) Using Superposition theorem, calculate the current flowing in the 200Ω resistor.

(d) Obtain the current flowing in each branch by using Kirchhoff's Laws.



Q2 (Total 20 Marks)

(a) A d.c. electric motor consumes $36\ \mathrm{MJ}$ when connected to a $250\ \mathrm{V}$ supply for one hour. Find: (i) the **power rating** of the motor (ii) the **current taken** from the supply (iii) the **cost for electricity** to operate the motor for six hours. Assume cost per unit of electricity is $\text{Rs. } 20/-$. (b) A Bakelite capacitor is to be constructed to have a **capacitance** of $0.04\ \mu\mathrm{F}$ and to have a steady working potential of $1\ \mathrm{kV}$ maximum. Allowing a safe value of field stress of $25\ \mathrm{MV/m}$, find: (i) the **thickness of Bakelite** required (ii) the **area of plate** required if the relative permittivity of Bakelite is 5 (iii) the **maximum energy stored** by the capacitor (iv) the **average power developed** if this energy is dissipated in a time of $20\ \mu\mathrm{s}$. (c) In a two conductor transmission system of $66\ \mathrm{kV}$, conductors are each of diameter $0.84\ \mathrm{cm}$ and the spacing between them $2.25\ \mathrm{m}$. Calculate the **capacitance per kilometer of line**.

Q3 (Total 20 Marks)

(a) Define the **reluctance** of a magnetic circuit. (b) State the **relationship** between the **magneto-motive force** and the **reluctance** of a magnetic circuit. (c) A mild steel closed magnetic circuit has a mean length of $75\ \mathrm{mm}$ and a cross-sectional area of $320.2\ \mathrm{mm}^2$. A current of $0.40\ \mathrm{A}$ flows in a coil wound uniformly around the circuit and the flux produced is $200\ \mu\mathrm{Wb}$. If the relative permeability of the steel at this value of current is 400 and $\mu = 4\pi \times 10^{-7}$, find: (i) the **reluctance** of the material (ii) the **number of turns** of the coil. (d) A flux of $30\ \mathrm{mWb}$ links with a $1200\ \mathrm{turn\ coil}$ when a current of $5\ \mathrm{A}$ is passing through the coil. Calculate: (i) the **inductance** of the coil (ii) the **energy stored** in the magnetic field (iii) the **average e.m.f. induced** if the current is reduced to zero in $0.20\ \mathrm{s}$.

Q4: Electromagnetism and AC Series Circuits (Total 20 Marks)

(a) A $25\ \mathrm{cm}$ long conductor moves at a uniform speed of $8\ \mathrm{m/s}$ through a uniform magnetic field of flux density $1.2\ \mathrm{T}$. Determine the **current flowing in the conductor** when: (i) its ends are **open-circuited** (ii) its ends are connected to a **load of $15\ \Omega$ resistance**. (b) A coil of inductance $159.2\ \mathrm{mH}$ and resistance $20\ \Omega$ is connected in series with a $60\ \Omega$ resistor to a $240\ \mathrm{V},\ 50\ \mathrm{Hz}$ supply. Determine: (i) the **impedance** of the circuit (ii) the **current** in the circuit (iii) the circuit **phase angle** (iv) the **p.d. across the $60\ \Omega$ resistor** (v) the **p.d. across the coil** (vi) Draw the **phasor diagram**.

Q5 (Total 20 Marks)

(a) State the **advantages of three-phase systems over single-phase system**. (b) **Compare star connected three-phase system with delta connected three-phase system**. (c) Each phase of a **delta-connected load** comprises a **resistance of $30\ \Omega$ and an $80\ \mu\mathrm{F}$ capacitor in series**. The load is connected to a $400\ \mathrm{V},\ 50\ \mathrm{Hz},\ 3\text{-phase supply}$. Calculate: (i) the **phase current** (ii) the **line current** (iii) the **total power dissipated** (iv) the **kVA rating** of the load. *Draw the complete **phasor diagram** for the load*. (d) Define the following terms used in **electrochemistry**: (i) **Electrolysis** (ii) **Polarization** (iii) **Corrosion** (iv) **Electroplating**

Q6 (Total 20 Marks)

(a) A **toroid** is of $30\ \mathrm{cm}$ mean circumference length and cross-sectional area $5\ \mathrm{cm}^2$. It is to be used as a magnetic core. **Two coils** having $100\ \mathrm{turns}$ and $28\ \mathrm{turns}$ are uniformly wound on the magnetic core having a **relative permeability of 800**. Assume that the **coupling coefficient is** 0.8. Determine: (i) the **inductance of each winding** (ii) the **mutual inductance**. (b) A **capacitor C** is connected in series with a $40\ \Omega$ **resistor** across a supply of **frequency** $60\ \mathrm{Hz}$. A **current of** $3\ \mathrm{A}$ flows and the **circuit impedance is** $50\ \Omega$. Calculate: (i) the **value of capacitance, C** (ii) the **supply voltage** (iii) the **phase angle** between the supply voltage and current (iv) the **p.d. across the resistor** (v) the **p.d. across the capacitor** (vi) Draw the **phasor diagram** for the above circuit.